TIANCHUN WU | 武天淳

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EDUCATION

THE CHINESE UNIVERSITY OF HONG KONG — **Hong Kong**
Bachelor of Engineering in Electronic Engineering, ELITE Stream — *Aug 2024 - Dec 2027*
CGPA: 3.37/4.0 | Advanced Standing | Coursework: Data Structures, C Programming, Digital Circuits, Electronic System Design

HARBIN INSTITUTE OF TECHNOLOGY — **Harbin, China**
HIT-CUHK Joint International Winter School — *Dec 2025*
Coursework: Introduction to Robotics; Introduction to Autonomous Unmanned Rotorcraft Systems

RESEARCH EXPERIENCE & SELECTED PROJECTS

RPAI LAB, THE CHINESE UNIVERSITY OF HONG KONG — **Hong Kong**
Research Student Helper | Supervisor: Mr. Chi Kit Ng — *Dec 2025 - Present*

- ColonVLA — Developing a motor-based mixture-of-experts Vision-Language-Action framework for autonomous colonoscopy withdrawal, using latent action-mode decomposition to model inspection, tracking, sampling, recovery, and anatomy-dependent control across a six-actuator endoscope platform.
- Integrated endoscopic video, motor states, clinical phase/anatomy labels, and electromagnetic tracking; used 3D tracker coordinates to estimate withdrawal velocity and support synchronized multimodal training and phantom-based validation.
- EndoNav (formerly StenoVLA) — Designed a supervised autonomous endoscopic navigation workflow in which human instructions interrupt autonomous control and assign the operative instrument, while a binary verifier governs safe continuation or termination during phantom experiments.
- EndoVLA-OOV / OpenHDS — Collected and structured 3,000+ endoscopic samples from anatomical phantoms for VLA development; contributed to the EndoVLA-OOV work submitted to TRL in collaboration with NVIDIA.
- Implemented research infrastructure spanning STM32 and CAN motor control, ROS/ROS 2, Python control interfaces, YOLO-assisted labeling, synchronized CSV/video acquisition, and six-motor robot-state logging.

ADDITIONAL RESEARCH PROJECT & AWARD

EXPLORING CONTINUUM ROBOT AGENTS FOR FIRE RESCUING IN BUILDINGS — **Hong Kong**
Project Team Member | Continuum Robotics and Agentic Autonomy — *2025 - 2026*

- Developed a continuum-robot agent concept for navigation and rescue assistance in structurally constrained building-fire environments, integrating robotic mobility, perception, and task-level autonomy.
- Received an award at the Hong Kong University Student Innovation and Entrepreneurship Competition.

PROFESSIONAL EXPERIENCE

TCL TECHNOLOGY CORPORATION — **Huizhou, China**
Project Assistant, Smartphone R&D — *Jun 2025 - Jul 2025*

- Supported cross-functional smartphone development by preparing technical documentation, executing 20+ functional tests across three prototypes, and reporting 8+ critical issues for engineering triage.
- Improved defect traceability and review coordination, contributing to a 10-day acceleration of the flagship-model iteration cycle.

LEADERSHIP & TECHNICAL ACTIVITIES

CHUNG CHI INTERNATIONAL ASSOCIATION — **Hong Kong**
President — *Apr 2025 - Present*

- Lead a 15-member executive team; delivered 10+ events for 1,000+ participants, secured two funding partners, and launched a mentorship program for 50+ incoming international students.

ROBOMASTER TEAM, THE CHINESE UNIVERSITY OF HONG KONG — **Hong Kong**
Member, Vision & Algorithms Group — *Dec 2025 - Present*

- Develop computer-vision capabilities for robotic platforms, focusing on real-time object detection, target tracking, and deployment-oriented model optimization.

TECHNICAL SKILLS

Programming: Python, C, C++
Robotics & Embedded: STM32, CAN, ROS/ROS 2, Linux, electromagnetic tracking, embedded-system design
Computer Vision & AI: YOLO, object detection, visual tracking, Vision-Language-Action models, multimodal data pipelines
Languages: Mandarin (native), Cantonese (fluent), English (fluent; IELTS 7.0)